ResearchHorizon — Updated Plan (v2)

Multi-Agent Future-Research-Direction Recommendation with MRR-Calibrated Tensor Rank Fusion

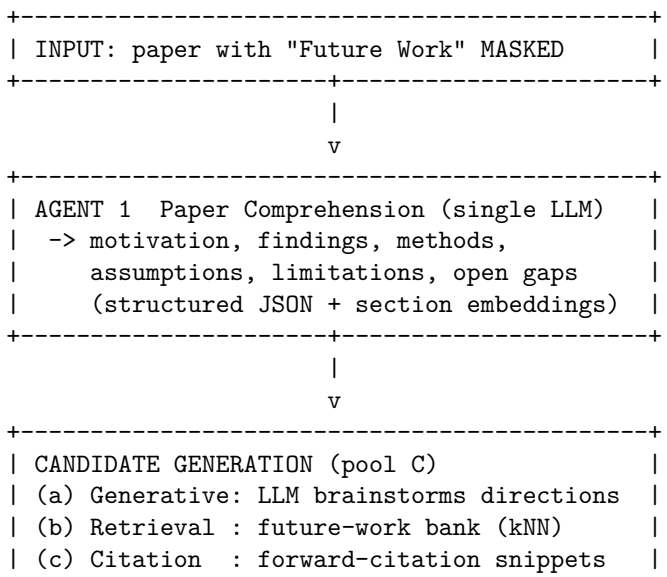
Use case. Given a paper with its "Future Work" / "Future Directions" section masked, recommend a *ranked list* of future research directions. The held-out future-work text is the ground truth, and the whole system is optimized for **Mean Reciprocal Rank (MRR)** of the true direction(s).

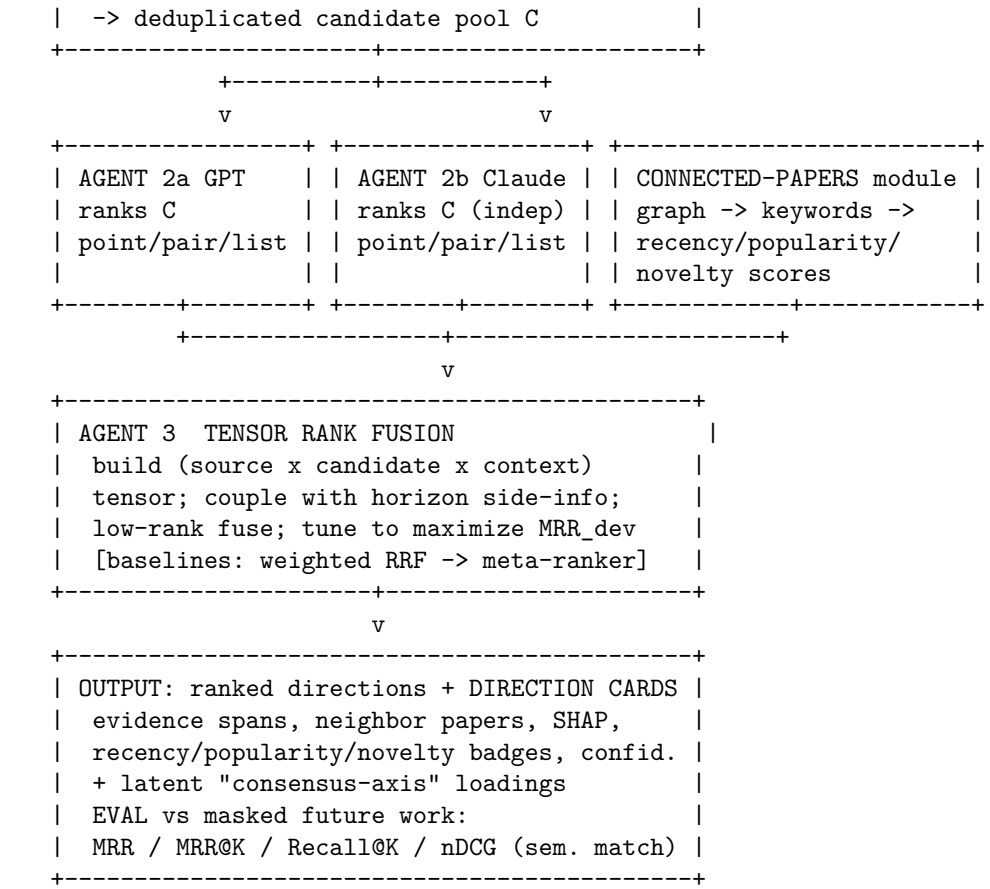
Design principle (the whole plan follows from this). A comprehension LLM reads the paper and produces a structured understanding; GPT and Claude each generate a ranking of candidate directions; the final ranking is decided by *validation MRR via rank fusion* — not by any single LLM's judgment. Crucially, every recommended direction ships with a transparent, evidence-grounded rationale (which limitation it addresses, which neighbor papers support it, which signals drove its rank), so the output is auditable, not a black box. This keeps the MLRC-Bench spirit: an objective metric, masked ground truth, real ablations, a compute-aware design — plus a model-risk-aware explainability layer.

One-line pitch. "We mask a paper's future-work section, have two LLMs independently rank candidate directions, fuse their rankings into a single list calibrated directly to dev MRR — and explain every recommendation with the paper evidence and a recency / popularity / novelty score from the citation graph."

> **What changed in v2.** The fusion stage (Agent 3) is upgraded from scalar-weighted RRF to a **context-conditioned low-rank tensor fusion** that (i) lets per-source trust adapt to the paper, (ii) captures ranker-agreement interactions with an explicit low-rank term, and (iii) couples the ranking tensor with the citation-graph horizon signals (CMTF). Scalar RRF and the learned meta-ranker are retained as the baselines the tensor model must beat. See the new **Section 5** and the **Novelty & Positioning** section (Section 12).

0. Flowchart





1. Task, data, and ground truth

Task. Input = paper body with the future-work section removed. Output = ranked list of K candidate future-research directions (short statements). The system never sees the answer at inference.

Ground truth = the masked future-work statements. Split each section into atomic directions $g_1, \ldots, g_m$ (one sentence/clause each). A direction can also be *weakly confirmed by forward citations* (papers that cite this one and actually pursued a direction) — a second, time-grounded label source.

Data sources. arXiv + ACL Anthology full text; S2ORC / unarXive (already section-parsed, so the future-work block is easy to isolate); Semantic Scholar API for the citation graph (forward citations). Keep only papers that contain an explicit future-work section.

Splits (no leakage). *Temporal split* by publication date: train/dev on older papers, test on the most recent — so candidate banks and any tuned weights cannot peek at future knowledge. Select everything on dev, report once on test.

How MRR is computed on free text (key design point). Recommendations and ground truth are text, not discrete IDs, so we need a match function $\text{match}(d_i, g_j) \in \{0, 1\}$:

```
match(d_i, g_j) = 1 if sim(d_i, g_j) >= tau            # embedding cosine threshold
                  or entailment_judge(d_i, g_j) == "covers"
```

For each true direction g_j: $\text{rank}_j = \min\{i : \text{match}(d_i, g_j) = 1\}$ (∞ if no match in top-K); $RR_j = 1/\text{rank}_j$ (0 if none); Paper MRR $= \text{mean}_j\, RR_j$. Corpus MRR = mean over papers of Paper MRR.

`sim` = cosine over SPECTER2 / SciNCL / SciBERT / sentence-transformers embeddings; τ is calibrated on a small human-labeled set. The entailment judge is an LLM used *only* for evaluation matching (report its agreement vs humans, e.g. Cohen's κ) — it never ranks.

2. Agent 1 — Paper Comprehension Agent

Role. Turn the masked paper into a compact, structured "understanding" that conditions everything downstream. No ranking, no future-work guessing here.

Output schema (structured JSON).

```
{ "problem": "...", "motivation": "...", "contributions": ["..."],
  "methods": ["..."], "datasets": ["..."], "assumptions": ["..."],
  "limitations": ["..."], "scope_boundaries": ["..."],
  "unexplored_axes": ["data","scale","modality","theory","evaluation","deployment"],
  "key_terms": ["..."], "embedding_text": "concatenated query text for retrieval" }
```

Methods to choose from. Long-context single pass (Claude/GPT, 1M-context) vs map-reduce / refine summarization for very long papers (chunk -> per-chunk notes -> merge); section-aware extraction (feed Intro / Method / Results / Limitations separately — they carry the most signal for "what's next"); scientific embeddings for the retrieval query (SPECTER2, SciNCL, SciBERT, or all-mpnet-base-v2); self-consistency (sample the summary n times, keep majority-agreed fields to reduce hallucinated assumptions).

Loss functions. Frozen LLM (default MVP, no training loss); if fine-tuned, token-level cross-entropy / SFT on (paper -> structured summary) pairs; optional DPO on pairs where one summary led to higher downstream MRR (aligns Agent 1 to the end metric); embedding model trained with contrastive InfoNCE / multiple-negatives ranking loss.

3. Candidate generation (the pool the rankers see)

The rankers can only reorder candidates they are given, so *candidate recall is the ceiling* (see 8.2). Build the pool from three complementary sources:

- **(a) Generative (LLM brainstorm).** Prompt GPT and Claude from Agent 1's summary to propose candidate directions ("list plausible next research steps; be specific; do not rank"). High recall, some noise.
- **(b) Retrieval (future-work bank).** Pre-index the future-work statements of all training papers; retrieve the kNN statements nearest the query embedding. Grounds candidates in what the field actually did. FAISS/HNSW kNN over SPECTER2/SciNCL; optional BM25 hybrid.
- **(c) Citation-grounded (optional, strong).** Pull sentences from forward-citing papers' intros ("Unlike [this paper], we …") — these often state the realized next step.

$C = \text{dedup}(C_{\text{gen-gpt}} + C_{\text{gen-claude}} + C_{\text{retrieval}} + C_{\text{citation}})$, ~50–200 candidates.

Loss functions (only the retriever trains). Contrastive InfoNCE / NT-Xent, triplet, or multiple-negatives ranking loss to pull a paper toward its true future-work statements; hard-negative mining (same topic, wrong specifics).

4. Agents 2a & 2b — GPT & Claude Rankers

Each LLM independently ranks the candidate pool C given Agent 1's summary. Two different model families = two semantic views; the value is *diversity of error*, not a majority vote. Output per agent: ranked list + per-candidate score + brief rationale.

Paradigm	How	Cost	Notes
Pointwise	score each candidate 0–1 for "is this the next step"	low (parallel)	simplest; loses cross-item context
Pairwise (PRP)	ask "which of A, B is more likely?"; aggregate wins	medium-high	strong but $O(n^2)$; use sorting / k-tournaments

Paradigm	How	Cost	Notes
Listwise (RankGPT)	rank a window in one call; sliding window for long lists	medium	best quality/cost trade-off
Setwise	pick top-k from a set per call	medium	fewer calls than pairwise

Aggregation within an agent. For pairwise, convert wins to a score with Bradley-Terry / Elo; for listwise sliding windows, stitch with overlap. Add self-consistency (multiple samples -> average rank).

Loss functions (only if you fine-tune / distill; otherwise prompt-only). Pointwise: BCE or MSE on graded relevance. Pairwise: RankNet, margin/hinge (RankSVM), LambdaRank. Listwise: ListNet, ListMLE, Softmax/ListCE, ApproxNDCG / NeuralNDCG, LambdaMART for GBDTs. Metric-aligned tuning: RLAIF / DPO with preferences from which ranking produced higher RR, or REINFORCE with MRR as reward (MRR is non-differentiable, so it enters as a reward). Cheaper: distill the LLM ranking into a cross-encoder reranker (SciBERT/MiniLM).

5. Agent 3 — Tensor Rank Fusion & MRR optimization

This is the heart of v2. The fusion problem is: given, for each paper p, a set of ranked lists from heterogeneous sources $m \in \mathcal{M} = \{\text{gpt}, \text{claude}, \text{retrieval}, \text{citation}, \text{horizon}\}$, produce one ranking calibrated to dev MRR. We describe the two retained baselines, then the tensor model that generalizes both.

5.1 Baseline A — Weighted Reciprocal Rank Fusion (RRF)

Robust, no training, strong baseline:

$$\text{RRF}(d \mid p) = \sum_{m \in \mathcal{M}} w_m \, \phi_m(d \mid p), \qquad \phi_m(d \mid p) = \frac{1}{k_m + r_m(d \mid p)}$$

where $r_m(d \mid p)$ is d's rank from source m (term dropped when m did not rank d), w_m is a *global scalar* weight, and k_m is per-source smoothing (start $k = 60$). Tune (k, w) to maximize dev MRR via grid / Bayesian search (Optuna/TPE). Start $w_{\text{gpt}} = w_{\text{claude}} = 1.0$, $w_{\text{retrieval}} = w_{\text{citation}} = 0.5$.

Limitations that motivate the tensor. (i) The weights are **global** — they cannot say "trust retrieval more on short, single-dataset papers." (ii) The model is **additive** — it cannot reward the fact that GPT and Claude *agreeing* is worth more than either alone. (iii) The horizon signals are bolted on as just another additive source.

5.2 Baseline B — Learned meta-ranker

One row per candidate, label = matches a true direction; features below. Model: LightGBM `LGBMRanker` (group = paper), logistic, or a cross-encoder; trained with an LTR loss (LambdaRank / pairwise logistic / softmax), **checkpoint and hyperparameters selected by dev MRR**.

Feature family	Examples
Rank-fusion	`rank_gpt, rank_claude, rank_retrieval, rrf_score, best_rank, mean_rank, rank_std, num_sources`
Agreement	`gpt_claude_rank_corr, both_top10_flag, disagreement_score`
Semantic	`sim_to_paper, sim_to_limitations, novelty_vs_related_work, topic_match`

Feature family	Examples
LLM signal	`gpt_score, claude_score, gpt_confidence, claude_confidence, rationale_length`
Provenance	`from_generative, from_retrieval, from_citation, retrieval_neighbor_dist`
Paper-level	`paper_length, #limitations, field, year, #contributions`

5.3 The tensor formulation (v2 contribution)

Assemble the fusion tensor. Stack the reciprocal-rank signal into a 3-mode tensor

$$\mathcal{X} \in \mathbb{R}^{P \times D \times M}, \qquad \mathcal{X}[p, d, m] = \phi_m(d \mid p)$$

(0 where source m did not rank d for paper p). Weighted RRF is exactly the operation that *collapses mode M of this tensor with a fixed weight vector w*. The tensor view keeps that structure and lets the fusion adapt.

(A) Context-conditioned low-rank weights (the primary, inductive model). Let $z(p) \in \mathbb{R}^F$ be paper-context features (field, length, #limitations, GPT–Claude agreement stats — the "Paper-level" and "Agreement" families above). Replace the global weight w_m with a context-dependent weight generated by a **rank-R factorization**:

$$w_m(p) = \langle U_m,\, z(p) \rangle, \qquad U = [U_1; \ldots; U_M] \in \mathbb{R}^{M \times F},\ U = AB^\top,\ \text{rank}(U) = R$$

$$s(d \mid p) = \sum_{m \in \mathcal{M}} w_m(p)\, \phi_m(d \mid p) = \sum_m \langle U_m, z(p) \rangle\, \phi_m(d \mid p).$$

This is a **bilinear / low-rank tensor model coupling (source $\times$ context $\times$ rank-signal)**. It *reduces to RRF* when U_m is constant in z, and it generalizes query-dependent scalar weighting (which is unregularized and rank-unconstrained) by sharing an R-dimensional latent structure across sources and contexts — crucial when papers are few.

(B) Source-interaction term (factorization-machine view). To capture agreement explicitly, add low-rank second-order terms:

$$s_2(d \mid p) = \sum_{m < m'} \langle v_m, v_{m'} \rangle\, \phi_m(d \mid p)\, \phi_{m'}(d \mid p), \qquad v_m \in \mathbb{R}^{R'}.$$

The GPT$\times$Claude agreement now emerges as a *learned multiplicative term* (a symmetric tensor factorization / FM), rather than a hand-crafted "agreement" feature.

(C) Coupled matrix-tensor factorization (CMTF) with the citation graph. Couple $\mathcal{X}$ (sharing the candidate mode D) with the horizon side-information matrix $H \in \mathbb{R}^{D \times 3}$ (recency / popularity / novelty from Section 7):

$$\min_{A,B,C,W} \left\| \mathcal{X} - [\![A, B, C]\!] \right\|_F^2 \;+\; \lambda \left\| H - BW^\top \right\|_F^2 \;+\; \Omega(\cdot)$$

so the Connected-Papers signals live in the *same latent space* as the ranking consensus — a principled version of "horizon features as an extra source," and the coupling that most distinguishes this work from generic rank fusion.

Full-CP / Tucker variant (when sources scale). If $\mathcal{M}$ grows (multiple LLMs $\times$ prompting paradigms $\times$ embedding spaces), promote $\mathcal{X}$ to a genuine multi-mode tensor and use CP/PARAFAC or Tucker; the mode-M factor gives interpretable per-source loadings on each latent "consensus axis." *For the hackathon-scale data, default to the small-R context-conditioned model (A)+(B); CP/Tucker is the scale-up path.*

5.4 MRR calibration and fitting

MRR is non-differentiable (it depends on discrete ranks). Fit the tensor factors with a **surrogate LTR loss** (pairwise logistic / ListNet / ApproxNDCG / SoftRank) but **select R, R', λ, k_m, and the loss by true dev MRR** — identical to the meta-ranker's honest protocol. Optimization: alternating least squares or Adam on the factor matrices; grouped by paper. Report bootstrap CIs and a paired significance test vs the RRF and meta-ranker baselines (small N demands it).

5.5 Interpretability payoff

The context weights $w_m(p)$ and the latent factor loadings are directly readable and feed the Direction Cards (Section 6): "for this paper archetype (short, single-dataset), the *retrieval+citation consensus axis* dominates and the horizon axis is down-weighted." Latent factors become **named consensus axes**, augmenting the SHAP/attribution layer rather than replacing it.

Why the tensor might (and might not) beat the baselines — the hypothesis we test, honestly. *For it:* context-adaptive trust + explicit interaction terms with far fewer parameters than a GBDT, a low-rank inductive bias matched to the "few sources, few papers, want interpretable source-trust" regime. *Against it:* with only ~5 sources and a small paper set a full CP/Tucker is data-starved, and a well-tuned LightGBM meta-ranker may already capture the interactions. **The contribution stands only if the ablation (Section 10) shows a real marginal MRR lift over both baselines with the same features; if it does not, the paper's contribution is the task/eval framing, not the tensor.** We commit to reporting this either way.

6. Reasoning & explainability (the "why" behind every direction)

A ranked list is not enough — a researcher must trust and verify each suggestion. Every output direction ships as a **Direction Card** with a faithful, evidence-grounded rationale (the model-risk-aware layer), and the explanation is checked for faithfulness.

Layers of explanation. Evidence grounding (extractive: link each direction to the specific limitation / assumption / open gap from Agent 1, with quoted spans); agent rationales (shown, not trusted — displayed with agreement/disagreement flags, but they never decide the rank); provenance (which source surfaced the candidate, with concrete precedent papers); feature attribution (SHAP local + global gain for the meta-ranker; **factor loadings / consensus-axis weights for the tensor model**); counterfactual / ablation probe ("remove citation evidence -> drops to rank 7"); confidence & abstention (calibrated score, report ECE; "insufficient evidence" when thin); faithfulness check (verify the displayed rationale matches the actual drivers; report a rationale-faithfulness rate).

Direction Card (output schema).

```
{ "direction": "...", "rank": 1, "score": 0.81, "confidence": "high",
  "addresses": ["limitation L3", "assumption A2"], "evidence_spans": ["...quoted..."],
  "supported_by": ["neighbor:Paper X", "citation:Paper Y"],
  "agent_view": {"gpt_rank": 2, "claude_rank": 1, "agreement": "high"},
  "horizon": {"recency": 0.74, "popularity": 0.40, "novelty": 0.66},
  "consensus_axes": {"retrieval_citation": 0.62, "llm_agreement": 0.55, "horizon": 0.18},
  "top_factors": ["agent_agreement", "retrieval_support", "recency_trend"] }
```

7. Extension — Connected-Papers signals: recency / popularity / novelty

Idea. Use the paper's related-citation neighborhood (co-citation + bibliographic coupling, the kind connectedpapers.com builds) to mine keywords and attach three horizon priors to each candidate — feeding the tensor's coupled side-info matrix H (Section 5.3C) and the Direction Card as badges.

Practical note (be honest). connectedpapers.com has no official public API; reconstruct the neighborhood via the Semantic Scholar Graph API or OpenAlex (bibliographic coupling + co-citation). Connected Papers is the conceptual analog; the graph is the mechanism.

Pipeline. (1) Neighborhood: seed = the paper; build the related-papers graph, each neighbor p with year y_p, citations c_p. (2) Keywords: KeyBERT / YAKE / RAKE / TF-IDF or an LLM keyword pass, embedded. (3) Match: keyword overlap / embedding similarity of each candidate d to the neighborhood keyword set -> its topic cluster and matching neighbor set M_d. (4) Three scores (z-normalized across candidates within a paper):

```
popularity(d) = SUM_{p in M_d} log(1 + c_p)                 # citation mass (momentum)
recency(d)    = SUM_{p in M_d} exp( -(Y_now - y_p) / tau )   # or slope of yearly counts
novelty(d)    = adjacency(d, paper) * 1 / (1 + |M_d|)        # adjacent-but-sparse "white space"
horizon_prior(d) = a*z(recency) + b*z(popularity) + c*z(novelty)
```

`adjacency = cos(e_d, e_paper)` gates novelty so off-topic phrases are not called "novel."

How it enters the system. As the CMTF side-info matrix H (primary in v2); as an extra RRF/tensor source (rank candidates by `horizon_prior`); and as Direction-Card badges. **Caveats:** popularity is a bias signal, not truth — popular is *not* the same as what this paper will do — so it is a feature the model can down-weight, never a hard prior; novelty must be adjacency-gated; forward-citation realized directions remain the cleaner label. Always ablate the horizon features for their marginal MRR lift.

8. Evaluation

Primary metric: **MRR** (and MRR@5/10, Recall@K, nDCG@K, MAP) with the Section 1 match function.

8.1 Reporting. MRR overall and sliced by field, paper length, #ground-truth directions; report match-function sensitivity (sweep τ) so the score is not a threshold artifact.

8.2 Candidate-recall diagnostic (do this first). The rankers cannot fix a missing candidate:

$$\text{Recall@}K_{\text{cand}} = \frac{1}{N} \sum_{\text{papers}} \mathbf{1}\{\, \exists\, d \in \text{top-}K : d \text{ matches any true } g \,\}$$

Stage	Dev metric
Generative candidate Recall@50	–
Retrieval candidate Recall@50	–
Citation candidate Recall@50	–
Union candidate Recall@50	–
GPT ranker MRR	–
Claude ranker MRR	–
RRF MRR	–
Meta-ranker MRR	–
Tensor-fusion MRR	–

First show candidate sources lift recall, then show fusion lifts ordering.

8.3 Leakage control. Temporal split; build the future-work bank and embeddings from train only; for the meta-ranker and tensor use out-of-fold candidate/feature generation so behavioral/retrieval features do not see the answer.

9. Possible improvements (ranked by expected payoff)

1. **Forward-citation grounding** — citing papers as a second ground-truth signal and candidate source; strongest and most novel label angle.

2. **Cross-encoder reranker** distilled from the LLM rankers — cheap at inference, fine-grained matching.
3. **Diversity-aware re-ranking (MMR)** — penalize near-duplicate directions so top-K covers distinct gaps (directly helps MRR with several true directions).
4. **Disagreement as signal** — where GPT and Claude disagree most, route to a tie-break (pairwise PRP or the tensor's interaction term); spend compute where it matters.
5. **Compute-Risk Governor** (reuse existing asset) — a utility $U = (\text{expected MRR gain} \times \text{confidence}) / (\text{LLM cost} \times (1 + \text{risk}))$ decides per paper whether to run the expensive pass.
6. **Abstention / calibration** — "insufficient evidence" over low-confidence guesses; report ECE.
7. **Knowledge-graph / gap detection** — map the paper onto a concept graph, surface unexplored edges as candidates.
8. **Human-in-the-loop calibration** set for τ and the entailment judge (report inter-annotator agreement).
9. **RAG vs long-context ablation** — quantify whether retrieval beats pure long-context comprehension.

10. Ablation table (report this)

System	Candidate sources	Rankers	Fusion	Dev MRR
Retrieval-only baseline	retrieval	none	none	–
GPT ranker only	gen+retrieval	GPT	none	–
Claude ranker only	gen+retrieval	Claude	none	–
GPT + Claude	gen+retrieval	both	RRF	–
+ citation candidates	gen+retrieval+cite	both	RRF	–
+ horizon priors	all	both	RRF	–
+ learned meta-ranker	all	both	learned (GBDT)	–
+ tensor fusion (ctx weights)	all	both	**tensor (A)**	–
+ interaction term	all	both	**tensor (A+B)**	–
+ CMTF horizon coupling	all	both	**tensor (A+B+C)**	–
Full + governor + MMR + explanations	all	both	best	best

The decisive rows are the three tensor rows against `+ learned meta-ranker`: same features, only the fusion model changes. That isolates the tensor's marginal MRR lift — the empirical core of the paper.

Headline claim. "Two independent LLM rankers plus a retrieval/citation candidate bank improve candidate recall and ordering; a context-conditioned tensor fusion — coupled with citation-graph recency/popularity/novelty priors and calibrated on dev MRR, not LLM judgment — makes the final decision and adapts per-source trust to each paper; and every recommendation is explained with the paper evidence, its ranking drivers, and its latent consensus axes."

11. Two-day work plan (team split)

Three workstreams run in parallel and integrate as soon as the agent structure lands.

- **Nourah — agent scaffold.** Stand up Agents 1, 2a/2b, and an Agent 3 stub end-to-end on the prepared inputs. The integration backbone everyone plugs into.
- **Taniya — rank fusion (parallel, on synthetic data).** Build and validate the fusion metric/models — weighted RRF + MRR harness + learned meta-ranker, **and now the tensor-fusion model (Section 5.3)** — on synthetic rankings, ready to drop into Agent 3. The synthetic generator must produce the `(source x candidate x context)` structure and a coupled horizon matrix so the tensor path is exercised before real data arrives. Then move to the explainability layer (Section 6).
- **Naz — paper suggestions & extensions.** Connect paper suggestions to Semantic Scholar (citation graph / future-work grounding); build the recency/popularity/novelty priors (Section 7) that become the CMTF side-info.

Most defensible final method: Agent 1 comprehension -> (generative + retrieval + citation) candidates -> GPT & Claude listwise rankers -> **context-conditioned tensor fusion coupled with horizon priors (baselines: RRF, meta-ranker)** -> argmax MRR_dev, with evidence-grounded Direction Cards.

12. Novelty & Positioning (for the paper)

Honest bottom line: none of the individual building blocks is new — the contribution is their specific intersection, the task/eval framing, and the metric-calibrated, side-info-coupled tensor formulation. State this plainly; reviewers will find the priors otherwise.

Prior art the paper must cite and distinguish itself from.

- *Tensor / coupled factorization for fusion & recommendation* is established: Coupled Matrix-Tensor Factorization and Coupled Graph-Tensor Factorization with side information (Acar et al.; arXiv:1809.08353), tensor methods for multi-view / multi-index fusion in IR, and the CP/Tucker toolkit (Kolda & Bader). **=> "tensor rank fusion" alone is not novel.**
- *LLM-ranker ensembling* exists: LLM-Blender (PairRanker + GenFuser, arXiv:2306.02561), RRF in hybrid RAG, neural rank fusion over multiple sources. **=> "fuse multiple LLM rankers" alone is not novel.**
- *Query-/context-dependent and adaptive fusion weights* exist: personalization-aware ranking fusion, adaptive weighted fusion layers, query-dependent RRF weights. **=> "context-adaptive weights" alone is not novel.**
- *Future-work generation from papers* exists: FutureGen (LLM-RAG to generate a paper's future work, arXiv:2503.16561), and LLM research-ideation work. **=> the task itself has a precedent, but framed as free-form generation, not ranked recommendation.**

The defensible novel contribution (the intersection).

1. **Task / evaluation framing.** Future-work as a *masked, MRR-calibrated ranked recommendation* against held-out ground truth — with forward-citations as a second, time-grounded label — rather than free-form text generation scored by reference-overlap (as in FutureGen). An objective, harder-to-game benchmark framing on top of the MLRC-Bench recommendation task.
2. **Method.** A **context-conditioned low-rank tensor fusion of heterogeneous rankers** (two LLM families $\times$ retrieval $\times$ citation) **coupled (CMTF) with citation-graph scientometric side-information**, with rank / regularization / weights **selected directly by non-differentiable MRR**. To our knowledge this exact combination — tensor fusion of LLM rankers coupled with recency/popularity/novelty priors and calibrated to MRR, for scientific future-direction recommendation — has not been published.
3. **Interpretability.** Factor-loading "consensus axes" and per-paper source-trust weights folded into evidence-grounded, faithfulness-checked Direction Cards (a model-risk-aware explainability contribution).

What a reviewer will demand — and this plan pre-empts. (i) An ablation isolating the tensor's marginal MRR lift over both weighted RRF *and* the LightGBM/logistic meta-ranker with identical features (Section 10 decisive rows); if the tensor does not win, the honest contribution is the framing. (ii) Statistical significance (bootstrap CI / paired test) given small N. (iii) A dataset at meaningful scale — the temporal-split S2ORC/unarXive corpus with several sources per paper — so the tensor is not data-starved (current validation is synthetic + a single real ranker; a publication needs the real multi-source corpus built out). (iv) An explicit "primitives are known; contribution is integration + domain + calibration + coupling" statement.

Recommended venue framing. An applied / resource paper (e.g., a *Findings*, workshop on scholarly document processing / scientometrics, or a benchmark track) fits better than a pure methods venue, because the strength is the *task* + *integration* + *benchmark*, not a new tensor algorithm. If the tensor ablation shows a robust, significant lift, elevate the method claim; otherwise lead with the benchmark and the calibrated-fusion recipe.

References

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4. Cohan et al., *SPECTER* (and SPECTER2 / SciNCL) — citation-informed document embeddings.
5. Cao et al., *ListNet* 2007; Xia et al., *ListMLE* 2008; Burges, *RankNet* / *LambdaRank* / *LambdaMART*.
6. Lo et al., *S2ORC*; *unarXive* for section-parsed full text.
7. LightGBM `LGBMRanker`; XGBoost LTR (`rank:ndcg` / `rank:map`; MRR is not a LambdaMART objective — select by dev MRR).
8. Lundberg & Lee, *A Unified Approach to Interpreting Model Predictions* (SHAP), 2017.
9. Grootendorst, *KeyBERT*; Campos et al., *YAKE!* — keyword extraction.
10. Connected Papers (co-citation + bibliographic-coupling graph; no public API); Semantic Scholar Graph API / OpenAlex as the programmatic substitute.
11. Kolda & Bader, *Tensor Decompositions and Applications*, SIAM Review 2009 — CP/PARAFAC & Tucker.
12. Acar et al., *All-at-once Optimization for Coupled Matrix and Tensor Factorizations* (CMTF); *Coupled Graphs and Tensor Factorization for Recommender Systems* (arXiv:1809.08353).
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